Interval notation

1

a $(-1, 2]$ **b** $(-6, -3)$ **c** $[-3, \infty)$ **d** $(-3, \infty)$

e $(-\infty, 4)$

2

a $(\infty, -7)$ **b** $(-\infty, 6]$ **c** $[-4, \infty)$ **d** $[-7, 3]$

Domain and range

3

a

i Domain$= 8, 1, 3, 9, 2$

ii Range$= 5, 3, 4, 1, 9$

iii Function

b

i Domain$= 2, 3, 6, 8$

ii Range$= 3, 8, 1, 7, 2$

iii Not a function

c

i Domain$= 4, 8, 2, 7, 3$

ii Range$= 4, 8, 9, 1$

iii Function

d

i Domain$= 1, 6, 3, 8, 2$

ii Range$= 3, 2, 7, 1$

iii Function

e

i Domain$= 7, 8, 5, 3$

ii Range$= 1, 9, 3, 2, 6$

iii Not a function

f

i Domain$= 2, 4, 6, 8, 10$

ii Range$= 2, 4, 6, 8, 9$

iii Function

g

i Domain $= 3 \leq x \leq 7$

ii Range $= 5 \leq y \leq 8$

iii Function

h

i Domain $= 2 \leq x \leq 7$

ii Range $= 5$

iii Function

i

i Domain $= 5$

ii Range $= 2 \leq y \leq 6$

iii Not a function

j

i Domain $= 2 \leq x \leq 9$

ii Range $= 3 \leq y \leq 8$

iii Not a function

k

i Domain $= -7 \leq x \leq 7$

ii Range $= 0 \leq y \leq 7$

iii Function

l

i Domain $= -9 \leq x \leq 9$

ii Range $= -9 \leq y \leq 9$

iii Not a function

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m

i All real x

ii $y \geq 0$

iii Function

n

i $x \geq 0$

ii All real y

iii Not a function

4

a

i Domain $= (-\infty, \infty)$

ii Range $= (-\infty, 2]$

b

i Domain $= (-\infty, \infty)$

ii Range $= [-2, \infty)$

c

i Domain$= [0, 6]$

ii Range$= [0, 7]$

5

a $(-\infty, \infty)$ **b** $(-\infty, \infty)$ **c** $[-6, 6]$

6

a False **b** False **c** True **d** True

7

a 1 **b** No **c** All real x **d** $y > 0$

e $y = 128$ **f** $x = 8$

8

a 16 **b** $y \leq 16$ **c** All real x

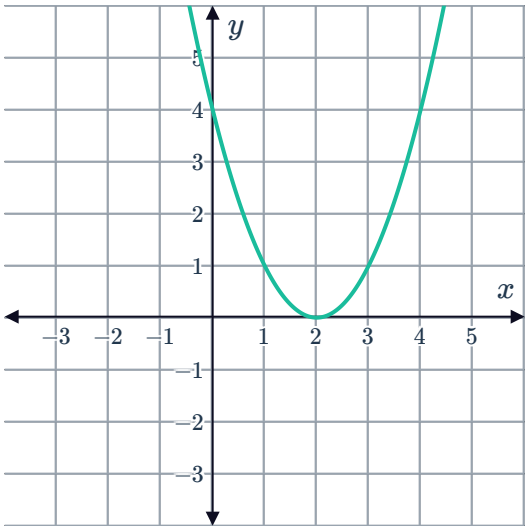
9

a $x \geq -1$ **b** No

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10

a



b No

Continuity

11

a Yes

b

i Yes **ii** Yes **iii** No

12

a Yes

b Yes

13

a Not continuous **b** Continuous **c** Not continuous **d** Continuous

e Continuous **f** Not continuous **g** Continuous **h** Not continuous

14

a False **b** True **c** False

15

a Yes **b** Yes **c** Yes

16

a Yes

b Yes

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17

a $(-\infty, 2) \cup (2, \infty)$ **b** Yes

c Yes

18

a $(0, \infty)$ **b** Yes **c** Yes

19

a No **b** $(-\infty, 2) \cup (2, \infty)$

20

a $[0, \infty)$ **b** $[0, \infty)$ **c** Yes

21

a $(-\infty, 5) \cup (5, \infty)$ **b** $(-\infty, 5) \cup (5, \infty)$

c Yes

22

a $(-\infty, \infty)$ **b** $(-\infty, \infty)$ **c** Yes

23

a $(-\infty, \infty)$

b $(-\infty, -5) \cup (-5, \infty)$

c No. The function is not continuous at every point in its domain.

24

a $(-\infty, \infty)$

b $(-\infty, 1) \cup (1, \infty)$

c No. The function is discontinuous at $x = 1$.

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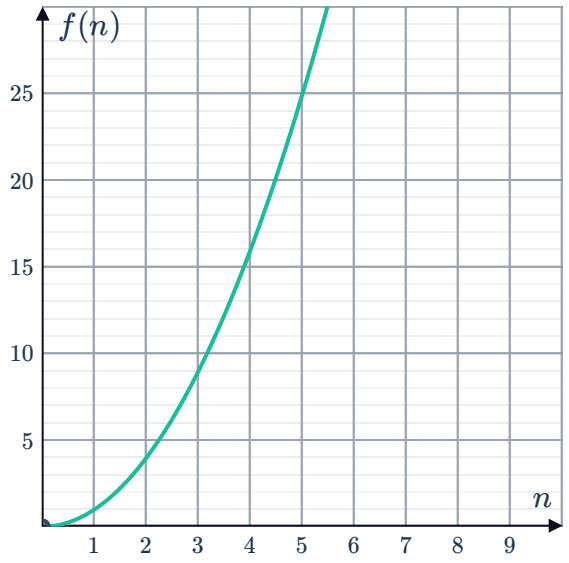
Applications

25

a **i** No **ii** Yes **iii** Yes **iv** Yes

b $f(n) = n^2$

c



26

a $0 \leq x \leq 7$

b 30

c No

27

a Yes

b The height of the water is a continuous function.

c We don't know because we don't know the location of the discontinuity.

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